

Practical Task 1

1a. The three countries with the lowest gender wage gap are Costa Rica (4%), Belgium (5%), and Denmark (6%).

1b. The three countries with the highest gender wage gap are Korea (37%), Japan (26%), and Israel (21%).

1c. Costa Rica was the country to have the lowest gender wage gap out of all the other countries in 2015. One reason why the gender wage gap is low is because of the country's strong commitment to gender equality. Costa Rica has laws that prohibit discrimination in employment and pay, resulting in employers being legally required to keep the gender wage gap low. Another reason why is because of the country's investments on education. There is a high educational attainment among women and thereby a greater participation in professions requiring a higher degree. Additionally, Costa Rica has had high female participation in the public sector, where the wage gap and pay scales are more publicized, reducing the wage differences in this sector. Furthermore, government policies that promote equal employment opportunities and access to healthcare and childcare have helped women stay in the workforce without having to leave for family reasons. Although there is still some wage gap, these factors likely contributed to Costa Rica having the lowest gender wage gaps among OECD countries in 2015.

2a. During March 2020, isopropanol prices shot up after being stable for most of the previous year. All three price lines have a sharp spike, showing a sudden and significant increase in the market price of isopropanol.

2b. A likely reason for the sharp spike could be COVID. Isopropanol is the key ingredient used in hand sanitizer and disinfectants, so the demand for isopropanol surged. However, manufacturers struggled to keep production matching with demand.

2c. The scatter plot shows a positive relationship between GDP per capita and CO2 emissions per person across all continents. Countries with higher income tend to produce more CO2 emissions per person.

In Africa, most countries are clustered at the lower end of GDP per capita and have lower CO2 emissions per person.

In the Americas, the countries spread across a wide range of income levels. Usually, CO2 emission increases as GDP per capita increases.

In Asia, the countries show great variation, with low-income, low-emission and high-income, high-emission.

In Europe, the countries are more concentrated at higher GDP per capita and moderate to high CO2 emissions. Some countries seem to have lower emissions.

In Oceania, they generally have high GDP per capita and high CO2 emissions per person.

Practical Task 2

1. The histograms show the distribution of each variable (Size of Living Area, Size of Garage, Year House Was Built, and Sale Price), indicating how frequently different values can occur.
2. Most garages are smaller than 1000 ft².
3. The most expensive houses were generally built after 1950.
4. There is a positive relationship between "Size of Living Area" and "Sale Price". As the size of living area increases, the sale price generally increases as well.